Γ-Net: The Minimum Reflection Principle and Three Inviolable Constraints

Framework Definition, Axiomatic Foundations, and Theorem Directory

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We present the complete axiomatic framework of the Γ-Net theory. The framework rests on a single first principle—the *Minimum Reflection Principle* (Γ-MRP): at every interface, the system acts to minimise total reflected energy. From this principle, three constraints emerge as a sequential deductive chain: C1 (energy conservation: $\Gamma^2 + T = 1$) follows from power balance; C2 (coefficient update: $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$) follows from C1 because persistent mismatch requires impedance adjustment; C3 (dimensional relay: K-staircase necessity) follows from C2 because cross-dimensional mismatch produces standing waves that C2 alone cannot resolve. C2 is proven to be the *unique* first-order activity-gated update rule that minimises reflected energy. We provide a complete inventory of derived quantities and a theorem directory mapping the deductive chain across five companion papers: Paper 1 (existence and irreducibility), Paper 2 (geometric quantification and metabolic scaling), Paper 3 (temporal unification and remodeling equivalence), Paper 4 (topological pathology), Paper 5 (full-system verification). This paper contains no new results beyond the derivation itself; it serves as the master reference for the series.

I. INTRODUCTION

The Γ-Net framework describes biological organisation as impedance physics. Every interface between two biological media—synapse, blood vessel wall, bone–cartilage junction, cell membrane, skin surface—is characterised by an impedance mismatch that determines how much energy is transmitted versus reflected.

Core physical narrative

Before entering the mathematical derivations, we summarise the entire theory in three physical pictures—one for each constraint. Every subsequent equation in this series is a quantitative expression of these three facts.

C1 — The thermodynamic imperative. A developing embryo is a high-metabolic heat source embedded inside a second heat source (the mother). Unless this heat is removed by convection, local temperature rises until proteins denature. Convection requires a pump; therefore the heart *must* be the first functional organ (human embryo: day 22). No genetic programme "chooses" this ordering: energy conservation (C1) demands it. The heart beats before the brain exists because there is no alternative consistent with the First Law of Thermodynamics (Paper 3, Remark **??**).

C2 — Phase separation creates Γ. To minimise internal reflection, cells with similar impedance cluster together—the physical process known as *phase separation*. But phase separation inevitably creates sharp boundaries between dissimilar phases (organs). At every such boundary, impedance mismatch ($\Gamma \neq 0$) is physically unavoidable. Life is therefore a system that, in building its own order, simultaneously creates the very friction it must spend its lifetime repairing. The coefficient update rule $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$ is the unique first-order algorithm for this perpetual repair (Section IV).

C3 — The quarter-wave bridge. When two phases have different modal complexities ($K_i \neq K_j$), C2 cannot reduce the mismatch because the receiving dimension does not exist ($\Gamma \to 1$). The only resolution is to insert intermediate relay tissues whose impedance is the geometric mean of the two extremes ($Z_{\rm relay} = \sqrt{Z_1 Z_2}$)—precisely the quarter-wave impedance transformer of microwave engineering. In the body, cartilage bridges bone to synovial fluid, the thalamus bridges cortex to brainstem, and arterioles bridge arteries to capillaries. These transition tissues are not biological curiosities; they are the physically mandatory solution to cross-dimensional reflection.

These three constraints—energy balance, impedance repair, and dimensional relay—are not biological peculiarities. They are consequences of energy transmission at interfaces, and would arise in *any* viable physical system with internal structure.

Framework overview

The framework rests on a single first principle—the Minimum Reflection Principle (Γ-MRP)—from which three constraints C1–C3 emerge as a sequential deductive chain, not independent postulates. This paper states the first principle, derives the three constraints, and proves the uniqueness of the central remodeling law (C2).

Five companion papers develop the consequences:

- **Paper 1** [1]: Proves that any finite network *must*

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have irreducible topological cost ($A_{\text{cut}} > 0$), derives the brain as the unique convergence structure, and extends to memory, consciousness, and nociception as impedance dynamics.

- **Paper 2** [2]: Quantifies the geometric cost: derives Murray's law, Kleiber's 3/4-power metabolic scaling, dual-network organ health, and the interface scale table spanning 14 orders of magnitude.

- **Paper 3** [3]: Proves C2 is algebraically equivalent to all 29 known tissue-remodeling laws, derives impedance debt, sleep, the lifecycle equation, aging, and emotion as temporal consequences of a single remodeling rule.

- **Paper 4** [4]: Derives disease as topological perturbation: E0 emergence of cardiovascular, neural, metabolic, and coupled multi-organ pathology from C1/C2/C3 alone, with zero disease-specific code.

- **Paper 5** [5]: Validates the full framework against NHANES epidemiological data (n = 49,774, zero outcome-fitted parameters), reports all-cause mortality AUC = 0.705, and collects all falsifiable predictions from Papers 1–4.

A reader entering at any single paper encounters local axiom recaps but lacks: (i) the derivation chain from Γ-MRP to C1/C2/C3; (ii) the proof of C2 uniqueness; (iii) a unified quantity inventory; (iv) a theorem directory mapping every major result to its location in the series. This paper fills those four gaps.

II. THE FIRST PRINCIPLE: Γ-MRP

A. Impedance at a one-dimensional interface

Consider any one-dimensional interface separating two media with generalised impedances Z_1 and Z_2, where impedance is defined as the ratio of effort (voltage, pressure, force, temperature) to flow (current, volume flow, velocity, heat flux):

$$Z_k = \frac{\text{effort}_k}{\text{flow}_k}\,. \tag{1}$$

Theorem 1 (Γ universality)**.** *At any one-dimensional interface between media of impedance Z_1 and Z_2, conservation of energy requires that incident power be partitioned into reflected and transmitted fractions:*

$$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}\,, \qquad T = 1 - \Gamma^2 = \frac{4Z_1Z_2}{(Z_1+Z_2)^2}\,, \tag{2}$$

where Γ is the reflection coefficient and T the transmission coefficient.

Proof. At a lossless one-dimensional interface, conservation of energy (First Law) requires: $P_{\text{incident}} = P_{\text{reflected}} + P_{\text{transmitted}}$. The boundary conditions on effort and flow (continuity of effort, conservation of flow) uniquely determine the reflected fraction as $|\Gamma|^2 = |(Z_2 - Z_1)/(Z_2 + Z_1)|^2$. This derivation applies identically to electrical transmission lines, acoustic waveguides, hydraulic conduits, mechanical interfaces, and thermal boundaries—the mathematical structure depends only on the existence of effort and flow variables, not on the physical medium. □

Theorem 1 is not a modelling choice. Any system with two impedances meeting at an interface *must* obey this partition. The reflection coefficient Γ is as fundamental as temperature or pressure: it exists wherever two impedances meet.

Remark 1 (Wave-independence of Γ)**.** *The reflection coefficient Γ and the power transmission $T = 1 - \Gamma^2$ are properties of* any *linear system with a source impedance and a load impedance. They do* not *require wave propagation. In an RF transmission line, Γ creates a reflected travelling wave; in a DC resistor circuit, Γ determines the power transfer efficiency without any wave; in an overdamped chemical system (e.g. insulin signalling), Γ quantifies the fraction of input signal that fails to activate the downstream pathway. The underlying theorem is the* maximum power transfer theorem *(Jacobi, 1840), which states that maximum power is delivered when $Z_{\text{load}} = Z^*_{\text{source}}$ ($\Gamma = 0$). This theorem holds for DC circuits, AC circuits, acoustic systems, hydraulic networks, and chemical flux systems—any domain where effort and flow variables are defined.*

Remark 2 (Dimensionless universality of Γ)**.** *Because $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$ is the ratio of two quantities with the same physical dimension, the units cancel identically:*

$$[\Gamma] = \frac{[Z] - [Z]}{[Z] + [Z]} = \frac{[Z]}{[Z]} = 1 \quad (\textit{dimensionless}).$$

This holds regardless of whether Z is measured in Ω (electrical), Pa·s/m^3 (vascular), N·s/m (mechanical), or any other effort/flow ratio. When the framework computes cross-domain quantities such as $H = \prod_i (1 - \Gamma_i^2)$ or $\Sigma\Gamma^2 = \sum_i \Gamma_i^2$, each Γ_i is already dimensionless, so the product and sum are well-defined without requiring transducer matrices or unit conversion. The cross-network coupling matrix C_{kj} (Papers 2–4) couples dimensionless Γ^2 values, not raw impedances.

B. The Minimum Reflection Principle (first principle)

For an organism with N interface channels, the total reflected (wasted) power is:

$$P_{\text{waste}} = \sum_{i=1}^{N} \Gamma_i^2\, P_{\text{in},i}\,. \tag{3}$$

An organism is *thermodynamically viable* if and only if

$$P_{\text{metabolic}} > P_{\text{waste}}\,, \tag{4}$$

i.e., its metabolic power production exceeds its reflection losses. Organisms that violate Eq. (4) die by energy deficit—no natural-selection argument is required.

Axiom 1 (Γ-MRP: the first principle)**.** *At every interface in a viable physical system, the total reflected energy over its operational lifetime is minimised:*

$$\boxed{\mathcal{A}[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t)\, P_{in,i}(t)\, dt \;\rightarrow\; \min\,.} \tag{5}$$

Γ-MRP is the single axiom of this framework. It is not derived from a deeper law; rather, conservation of energy at an interface (the thermodynamic First Law) is a *consequence* of Γ-MRP applied to a single time instant. The three constraints C1, C2, C3 are derived from Γ-MRP as a sequential deductive chain: C1 follows from instantaneous power balance; C1 generates C2 (persistent mismatch requires update); C2 generates C3 (cross-dimensional mismatch produces standing waves requiring relay insertion).

III. THE THREE CONSTRAINTS

A. C1: Energy Conservation

Theorem 2 (C1)**.** *At every interface i, at every instant t:*

$$\Gamma_i^2(t) + T_i(t) = 1\,, \tag{6}$$

where $T_i = 1 - \Gamma_i^2$ is the transmission coefficient.

Proof. By Theorem 1, at any lossless one-dimensional interface the reflected power fraction is $|\Gamma_i|^2$ and the transmitted fraction is $T_i = 4Z_1Z_2/(Z_1+Z_2)^2$. Direct substitution of $\Gamma_i = (Z_2 - Z_1)/(Z_2 + Z_1)$ yields:

$$\Gamma_i^2 + T_i = \frac{(Z_2-Z_1)^2}{(Z_2+Z_1)^2} + \frac{4Z_1Z_2}{(Z_2+Z_1)^2} = \frac{(Z_2-Z_1)^2 + 4Z_1Z_2}{(Z_2+Z_1)^2} = \frac{(Z_2+Z_1)^2}{(Z_2+Z_1)^2} = 1\,. \tag{7}$$

Dissipative corrections modify $Z_i(t)$ over time (Johnson–Nyquist noise, Arrhenius aging) but at each instant the power balance at the interface is governed by the instantaneous impedances, so C1 holds pointwise in time. □

C1 is Γ-MRP applied to a single interface at a single instant—the thermodynamic First Law is this special case. It holds exactly for any lossless interface. Dissipative corrections (Johnson–Nyquist noise, Arrhenius aging) modify Z over time but never violate power balance at a single interface.

Remark 3 (Signal power vs. metabolic work)**.** *C1 governs instantaneous* signal power *at each interface: incident = reflected + transmitted. The metabolic energy fuelling C2 update enters as a separate thermodynamic work term that modifies Z itself, not the $\Gamma^2 + T = 1$ identity. The analogy is an electronic amplifier: the DC power supply changes the gain (modifies Z), yet Kirchhoff's power law holds at every RF port.*

Corollary 1 (Thermal Dissipation)**.** *The reflected fraction Γ_i^2 of incident power is re-absorbed by the source impedance and dissipated as Joule heat:*

$$P_{\text{heat}} = \sum_{i=1}^{N} \Gamma_i^2\, P_{\text{in},i}\,. \tag{8}$$

The body temperature in thermal steady state satisfies $P_{\text{heat}} = hA\,(T_{\text{body}} - T_{\text{env}})$. The impedance mismatches that perform computation therefore generate the heat that sustains the temperature at which those computations remain viable.

B. C2: Coefficient Update (generated by C1)

C1 diagnoses mismatch ($\Gamma \neq 0$) but does not resolve it. For the system to obey Γ-MRP over time, it must *update* its impedances to reduce $\mathcal{A}$. This necessity generates C2:

Theorem 3 (C2 as gradient descent on $\mathcal{A}$)**.** *First-order gradient descent on the reflection action $\mathcal{A}$ (Eq. 5) with respect to Z_i, gated by input and output activity, yields:*

$$\boxed{\Delta Z_i = -\eta\, \Gamma_i\, x_{in,i}\, x_{out,i}\,,} \tag{9}$$

where $\eta > 0$ is the update rate, $x_{in,i}$ is the input signal, and $x_{out,i}$ is the transmitted signal.

Proof. The functional derivative of $\mathcal{A}$ with respect to Z_2 at an interface gives:

$$\frac{\delta \mathcal{A}}{\delta Z_2} = \frac{4Z_1(Z_2 - Z_1)}{(Z_2+Z_1)^3}\, P_{\text{in}} \propto \Gamma\,. \tag{10}$$

Gradient descent $\Delta Z = -\eta\,(\delta\mathcal{A}/\delta Z)$ with activity gating (update only when both input and output are active) yields Eq. (9). □

C2 is *not a postulate*: it is the unique first-order activity-gated update rule that minimises reflected energy (Section IV). Its form is identical to Hebb's learning rule when Z is synaptic conductance, to Wolff's law when Z is bone stiffness, to Glagov's remodeling when Z is vascular compliance, and to 26 further tissue-specific laws (Paper 3 [3]).

Historical note. In the original formulation, C2 was called the "Hebbian update" with signals x_{pre}, x_{post}. Paper 3 proves that the same equation governs all 29 tissue types, making neural-specific terminology misleading. We adopt *coefficient update* and $(x_{\text{in}}, x_{\text{out}})$ throughout.

C. C3: Dimensional Relay (generated by C2)

C2 adjusts impedance within a given dimension. But when two nodes have different modal complexities ($K_i \neq K_j$), the extra modes in the higher-K node produce standing waves with no counterpart at the lower-K node. C2 cannot resolve this mismatch because there is no Z to adjust—the receiving dimension does not exist. To satisfy Γ-MRP, the system must insert intermediate relay nodes, generating C3:

Theorem 4 (C3 (E0))**.** *If two nodes i, j have modal complexities $K_i \neq K_j$, then $\Gamma_{ij} = 1$ (total reflection). C2 cannot remove it because there is no Z-mismatch to minimise—the receiving dimension simply does not exist. The only resolution is to insert intermediate relay nodes forming a K-staircase $K_i > K_{m_1} > K_{m_2} > \cdots > K_j$, splitting the single $(K_i - K_j)$-mode total reflection into smaller reflections, each of which C2 can partially reduce.*

Proof. A mode in i with no corresponding degree of freedom in j sees infinite load impedance $\to \Gamma = 1$. This is topological, not impedance-based: C2 can only adjust Z within existing dimensions. By induction, a chain of relays $K_i > K_{m_1} > \cdots > K_j$ minimises total Γ^2 subject to the topological constraint. □ □

C3 explains why biological systems develop layered architecture:

- *Neural*: cortex ($K = 5$) → thalamus → brainstem → spinal cord → motor neuron → muscle ($K = 1$).
- *Vascular*: aorta → arteries → arterioles → capillaries—a fractal K-staircase for impedance matching.
- *Skeletal*: bone → cartilage → tendon → muscle—graded stiffness across the joint interface.

Without dimensional relay, high-K controllers and low-K effectors would be separated by 100% reflection walls, and complex life would be physically impossible.

Remark 4 (Metadata co-transport)**.** *Every energy or material flow across an interface carries (Z_{source}, Z_{load}) as metadata, so that Γ_i is locally computable without global network knowledge. This is not an additional physical law but a necessary consequence of Theorem 1: since Γ_i depends only on the local pair $(Z_{source,i}, Z_{load,i})$, each carrier (electrical signal, blood, mechanical load) must encode its source impedance upon departure.*

Remark 5 (Phase separation creates Γ)**.** *C2 gradient descent drives cells with similar impedance into spatial clusters (same-K clustering, Paper 1 [1]). This is the physical process known as* phase separation*: liver cells aggregate with liver cells, bone with bone, neural tissue with neural tissue. Phase separation minimises* intra-*phase reflection, but it simultaneously sharpens the* inter-*phase boundaries. At every such boundary, impedance mismatch is physically unavoidable:*

$$Z_{phase\,A} \neq Z_{phase\,B} \implies \Gamma_{AB} \neq 0\,.$$

The irreducible topological cost $A_{cut} > 0$ (Paper 1, Theorem 2) is therefore not a design flaw but a thermodynamic consequence *of structural order. Life exists in a perpetual tension: C2 builds order by clustering, but clustering creates the very boundaries that generate Γ. The organism spends its entire lifetime running C2 to repair friction that C2's own success created. This self-referential loop is the physical reason why A_{cut} can never reach zero in any finite network.*

Implementation note. In the reference implementation (ALICE), C3 is enforced by requiring all inter-module values to be `ElectricalSignal` objects carrying Z metadata. Raw floating-point numbers between modules are forbidden.

IV. MATHEMATICAL UNIQUENESS OF C2

Theorem 5 (C2 uniqueness)**.** *Among all first-order, local, activity-gated update rules of the form $\Delta Z = f(Z, x_{in}, x_{out})$, the rule $\Delta Z = -\eta\,\Gamma\,x_{in}\,x_{out}$ is the* unique *descent direction that minimises $\mathcal{A}[\Gamma]$ at every step.*

Proof. Any first-order descent must follow $\Delta Z \propto -\delta\mathcal{A}/\delta Z$. By Eq. (10), $\delta\mathcal{A}/\delta Z \propto \Gamma$. Activity gating restricts updates to epochs where both $x_{\rm in} \neq 0$ and $x_{\rm out} \neq 0$. The unique factorisation satisfying locality, activity gating, and descent is $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$ with $\eta > 0$. □

A. The tissue is not the algorithm

The molecular machineries implementing C2 differ vastly across tissues: NMDA receptors (neural), RANKL/OPG (bone), eNOS (vascular), AID somatic hypermutation (immune), keratinocyte Ca^{2+} (skin), HGF (liver), EPO (marrow), FSH receptors (reproductive)—completely different molecular pathways implementing the same equation. The substrate determines the material constants (η, Z_0); the equation is dictated by physics.

B. Objective function for viability

The Γ-framework supplies a *physical* objective function: $\mathcal{A} = \int_0^T \sum_i \Gamma_i^2\,P_{{\rm in},i}\,dt$. Any system whose $\mathcal{A}$ exceeds the viability bound is physically excluded. This is a constraint on what *can* exist, not an explanation of what *does* exist.

C. Sex as a boundary condition

Sex does not alter C1–C3. The hypothalamic–pituitary–gonadal (HPG) axis sets sex-specific Z_{setpoint} values: testosterone raises η_{muscle}; oestrogen modulates Z_{adipose} and Z_{bone} (menopause $\equiv$ $Z_{\text{bone}} \uparrow$, $\Gamma_{\text{bone}} \uparrow$); pre-menopausal oestrogen maintains lower Γ_{vasc} via enhanced endothelial C2. Sex is a parametric shift in boundary conditions, not a different dynamical law.

V. DERIVED QUANTITIES

Table I lists every named quantity in the Γ-Net framework, ordered by the paper where it first appears.

VI. THEOREM DIRECTORY

The five companion papers form a deductive chain radiating outward from C1/C2/C3. Each paper occupies a unique mathematical role. Table II provides the complete map.

A. The deductive chain

The logical structure is a concentric expansion:

Layer	Content	
0	Γ-MRP → C1 → C2 → C3	(this paper)
1	C2 on finite network → $A_{\text{cut}} > 0$ → topology, brain, mind	(Paper 1)
2	C2 on branching geometry → Murray, Kleiber, H	(Paper 2)
3	C2 ≡ 29 laws → D_Z, sleep, lifecycle, aging	(Paper 3)
4	Topology + perturbation → disease emergence (E0)	(Paper 4)
5	Γ-vector + NHANES → validation, blueprints, predictions	(Paper 5)

Each layer uses only C1/C2/C3 plus the results of inner layers. No layer imports concepts from an outer layer. This unidirectional dependency is the structural guarantee against circularity.

VII. WHAT THIS THEORY DOES NOT CLAIM

The Γ-framework derives *physical constraints* that any living system must satisfy. It does *not* reconstruct thermodynamic history.

1. **Necessary conditions, not sufficient causes.** The framework shows that a K=5 brain *must* maintain $T > T_c(5)$ and *must* have vasculature. It does not explain *why* such brains appeared or *which* lineages developed them.

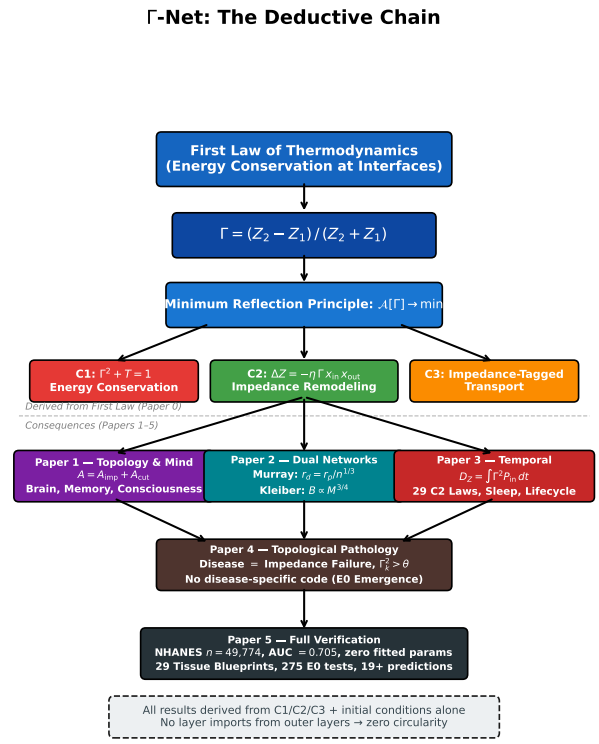

FIG. 1. The deductive chain of the Γ-Net framework. From the first principle Γ-MRP, the three constraints emerge as a sequential deductive chain; the Minimum Reflection Principle yields C1/C2/C3; each subsequent paper applies these constraints to a specific domain. No layer imports from an outer layer.

2. **No historical trajectory.** Statements such as "the tidal zone offers the lowest-action path" identify geometric optima in impedance space. Whether any lineage actually traversed that path is a question for palaeontology, not for this framework.

3. **C2 uniqueness is mathematical, not biological.** The proof that C2 is the unique first-order $\mathcal{A}$-minimising rule is a theorem about gradient descent. That multiple molecular machineries converge on the same equation is *consistent with* C2 being a physical attractor; it is not evidence that history "discovered" C2.

4. **Compatibility, not replacement.** Thermodynamics determines which lineages survive; the Γ-framework supplies the physical objective function that selection operates on.

Throughout the six papers, any sentence referencing the relationship to biology should be read as *physical con-*

TABLE I. Complete quantity inventory of the Γ-Net framework.

Symbol	Name	Definition	Interpretation	Paper
Γ_i	Reflection coefficient	$\frac{Z_{\text{load},i}-Z_{\text{source},i}}{Z_{\text{load},i}+Z_{\text{source},i}}$	Mismatch at interface i	0
T_i	Transmission coefficient	$1-\Gamma_i^2$	Fraction of power delivered	0
$\mathcal{A}$	Reflection action	$\int_0^T \sum_i \Gamma_i^2 P_{\text{in},i}\, dt$	Total wasted energy	0
A_{imp}	Improvable action	$\mathcal{A} - A_{\text{cut}}$	Reducible by C2 update	1
A_{cut}	Irreducible action	$\sum_{k>K^*} \Gamma_k^2$	Topological cost immune to C2	1
Φ_{meta}	Consciousness index	$1-\Gamma_{\text{meta}}^2$	Self-referential matching	1
$\mathcal{S}$	Invariant topological core ("soul")	$\ker(\partial Z/\partial t)$	Impedances immune to experience	1
β	Murray bifurcation ratio	$2^{1/3} \approx 1.26$	Optimal branch radius ratio	2
Ω	Dual-network overhead	≥ 1	Packing cost of two trees	2
H	Health index	$\prod_i (1-\Gamma_i^2)$	Whole-body transmission	2
D_Z	Impedance debt	$\int_0^{T_{\text{wake}}} \lvert\Gamma\rvert^2 P_{\text{in}}\, dt$	Cumulative mismatch; cleared by sleep	3
Lifecycle	Lifecycle ODE	$\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\, \Sigma\Gamma^2 + \gamma\, \Gamma_{\text{env}} + \delta\, D(t)$	Cognition bathtub curve	3
Π_Γ	Gamma-Pain Index	$w_n \Gamma_n^2 + w_v \Gamma_v^2 + w_D \frac{dD_Z}{dt}$	Consciousness-independent pain	4

straint language: "X is required" means "any system with property Y must satisfy X".

VIII. NOTATION CONVENTIONS

Table III collects notation used across Papers 0–5.

IX. CONCLUSION

The Γ-Net framework rests on a single first principle: the Minimum Reflection Principle (Γ-MRP, Axiom 1). From this axiom, three constraints emerge as a sequential deductive chain: C1 ($\Gamma^2 + T = 1$) from instantaneous power balance; C2 ($\Delta Z = -\eta\, \Gamma\, x_{\text{in}}\, x_{\text{out}}$) from the necessity to resolve persistent mismatch, proven unique among first-order activity-gated rules; C3 (K-staircase relay) from the standing waves that arise when C2 encounters cross-dimensional mismatch.

A. The minimal operational loop

The three constraints compress into a single three-step cycle that every interface must execute:

Constraint	Role	Physical content
C1	**Perceive**	$\Gamma^2 + T = 1$ partitions every signal into "reflected" and "transmitted." All sensation is a readout of this split.
C2	**Update**	$\Delta Z = -\eta\, \Gamma\, x_{\text{in}}\, x_{\text{out}}$ writes the perceived mismatch back into Z, changing how subsequent signals are partiti
C3	**Relay**	Dimensional relay propagates the local upd along the K-staircase to all connected node distributing the error across the network.

Every physiological phenomenon in Papers 1–5 is a manifestation of this *Perceive–Update–Relay* loop executing on different channels, at different timescales, with different material substrates.

Remark 6 (Perception × error × compensation)**.** *The three-step cycle admits a compact control-theoretic reading:*

1. ***Perception*** *(C1 readout): the organism estimates its current mismatch $\hat{\Gamma}^2$ from sensory channels.*
2. ***Error****: $e = \Gamma_{real}^2 - \hat{\Gamma}^2$.*
3. ***Compensation****: a behavioural or metabolic action u is selected to reduce e.*
4. ***Write-back*** *(C2): the compensation result is committed to Z, updating the next perception.*

TABLE II. Theorem directory: the deductive chain from C1/C2/C3 to all major results. Each row is a theorem or major result, with its paper of origin and the constraints required for its derivation. Theorems are grouped by paper and by mathematical role.

Paper	Role	Theorem / Result	Requires	Section
Paper 1: Existence and Irreducibility				
	Existence	$\mathcal{A} = A_{\text{imp}} + A_{\text{cut}}$, $A_{\text{cut}} > 0$	C1, C2, K-dim cascade	P1 §2–3
	Topology	Same-K clustering, relay hierarchy	$A_{\text{cut}} > 0$, C2	P1 §4–5
	Convergence	Brain as unique convergence structure	Relay hierarchy	P1 §6
	Memory	Memory $\equiv dZ/dt$ (three regimes)	C2	P1 §8
	Consciousness	$\Phi_{\text{meta}} = 1 - \Gamma^2_{\text{meta}}$; Television Theorem	C1, C2, feedback	P1 §9–10
	ITC ("Soul")	$\mathcal{S} = \ker(\partial Z/\partial t)$; four laws of null space	C2 uniqueness	P1 §11
	Nociception	Three-layer pain; Π_Γ index	C1, C2, SNR	P1 §13–14
Paper 2: Geometric Quantification				
	Vascular	Murray's law from Γ-minimisation; $\beta = 2^{1/3}$	C2, Poiseuille Z	P2 §2–4
	Scaling	$A_{\text{cut}} \propto M^{2/3}$; Kleiber $B \propto M^{3/4}$	A_{cut}, Murray, Ω	P2 §7–8
	Dual-network	$H = (1-\Gamma_n^2)(1-\Gamma_v^2)$; four failure modes	C1, C2, two buses	P2 §5–6
	Universality	Interface scale table (14 orders of magnitude)	Γ universality (Thm 1)	P2 §9
Paper 3: Temporal Unification				
	Equivalence	C2 $\equiv$ 29 tissue-update laws	C2, tissue substitutions	P3 §2
	Sleep	Sleep $= D_Z$ discharge; gate-leak inequality	C2, D_Z, Johnson–Nyquist	P3 §3–5
	Lifecycle	$\frac{d(\Sigma\Gamma^2)}{dt} = -\eta\Sigma\Gamma^2 + \gamma\Gamma_{\text{env}} + \delta D$	C2, D_Z, Arrhenius	P3 §6
	Aging	Coffin–Manson fatigue; death as $\Gamma^2_{\text{basal}} \geq 1$	C2, null space, D_Z	P3 §7
	Emotion	Dopamine = RPE; Yerkes–Dodson; curiosity	C2, D_Z	P3 §8
	Morphogenesis	Dual-field PDE ($\partial Z/\partial t$, $\partial\rho/\partial t$)	C2, reaction–diffusion	P3 §9
Paper 4: Topological Pathology				
	E0 cardio	HTN, MI, aortic stenosis from topology	C1/C2/C3, GammaTopology	P4 §2
	E0 neural	ALS, stroke, PTSD, phantom limb from topology	C1/C2/C3, GammaTopology	P4 §3
	Coupled	Five-organ cascade; three recovery regimes	C2, coupling matrix	P4 §4–5
	Metabolic	T2DM six-channel hexagonal deadlock	C2, bifurcation	P4 §6
	Necrosis	Tissue necrosis as substrate collapse	$D_Z > D^{\max}_{\text{repair}}$	P4 §7
	Sleep path.	Insomnia / narcolepsy / hypersomnia	Gate-leak inequality	P4 §8
Paper 5: Full-System Verification				
	NHANES	7/7 organs AUC significant; $C = 0.70$; 6.89× risk ratio	Γ universality, H	P5 §2
	Propagation	Cross-organ AUC gain (cardiac +0.166)	Network coupling	P5 §3
	Blueprints	29 tissue blueprints; 278 nodes, 426 edges	C1/C2/C3, K-cascade	P5 §4
	Ablation	C2-on/C2-off pincer tests	E0 emergence	P5 §5
	Predictions	19+ falsifiable predictions (unified list)	Papers 1–4	P5 §6

When compensation is correct, global Γ^2 decreases and perception recalibrates. When compensation is incorrect—reward misalignment, addictive substances, sensory distortion—the erroneous perception is written into Z by C2, creating a positive-feedback loop whose pathological consequences are derived in Papers 3 and 4.

From these three constraints, five companion papers derive, in a strictly outward deductive chain: topology and the mind (Paper 1), geometric quantification and metabolic scaling (Paper 2), temporal unification across 29 tissues (Paper 3), disease as topological pathology (Paper 4), and full-system verification against epidemiological data (Paper 5).

This paper provides the master reference: a derivation chain (Sections II–IV), a quantity inventory (Table I), and a theorem directory (Table II).

All source code, simulation scripts, and NHANES pipelines are available at `https://github.com/cyhuang76/alice-gamma-net` under AGPL-3.0.

[1] H.-Y. Huang, "The irreducibility theorem and the physics of mind: topology, memory, consciousness, and nociception from impedance constraints," companion Paper 1

TABLE III. Notation conventions used across all six papers.

Symbol	Meaning	Units / Range	Paper
Z	Impedance	Ω etc.	0
Γ	Reflection coefficient	$[-1, +1]$	0
T	Transmission $(1-\Gamma^2)$	$[0, 1]$	0
η	Update rate	> 0	0
$(x_{\text{in}}, x_{\text{out}})$	Interface signals	signal units	0
$\mathcal{A}$	Reflection action	J·s	0
K	Modal complexity	integer ≥ 1	1
A_{imp}	Improvable action	J·s	1
A_{cut}	Cutoff action	J·s	1
Φ_{meta}	Consciousness index	$[0, 1]$	1
$\mathcal{S}$	Null space	—	1
β	Murray ratio $(2^{1/3})$	dimensionless	2
Ω	Dual-network overhead	≥ 1	2
H	Health index	$[0, 1]$	2
Z_v, Z_n	Vascular / neural Z	domain-specific	2
D_Z	Impedance debt	J·s	3
$T_c(K)$	Critical temperature	°C	3
Π_Γ	Γ-Pain Index	dimensionless	4

(2026).

[2] H.-Y. Huang, "Geometric quantification: Murray's law, Kleiber scaling, and dual-network organ health from Γ-minimisation," companion Paper 2 (2026).

[3] H.-Y. Huang, "Temporal unification: C2 as twenty-nine tissue laws, impedance debt, sleep, and the lifecycle equation," companion Paper 3 (2026).

[4] H.-Y. Huang, "Topological pathology: disease emergence from C1/C2/C3 without disease-specific code," companion Paper 4 (2026).

[5] H.-Y. Huang, "Full-system verification: NHANES validation, tissue blueprints, and falsifiable predictions," companion Paper 5 (2026).

[6] H.-Y. Huang, *ALICE: A physics-driven electronic lifeform*, `https://github.com/cyhuang76/alice-gamma-net` (2026).